

AKASH SAXSENA

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PROFESSIONAL SUMMARY

Electronics Engineer (Research Development Engineer) with hands-on experience in designing, developing, and testing embedded hardware and microcontroller-based systems. Currently working at Dual Life Science Pvt. Ltd., with expertise in ESP8266, Embedded C, hardware-software integration, and communication protocols including I2C, SPI, and UART. Experienced in developing real-time embedded and IoT solutions, including RFID-based authentication, smart energy management, sensor interfacing, and web-integrated systems. Strong ability to troubleshoot hardware, integrate electronic components, and develop reliable solutions for RD applications.

WORK EXPERIENCE

NETSOL IT SOLUTION PVT. LTD.

02/2026 – 05/2026

Embedded Engineer Intern • Surat, Gujarat

- Engineered and deployed ESP8266-based IoT systems for smart automation and energy monitoring applications, enabling real-time device control and data tracking.
- Developed and integrated RFID-based authentication modules with multi-user access control, improving system security and usability.
- Designed and implemented hardware interfacing for sensors, RTC modules, and display units, ensuring reliable real-time system performance.
- Built and optimized firmware in Embedded C, enhancing system stability and reducing runtime errors during continuous operation.
- Implemented communication protocols (I2C, SPI) for seamless peripheral integration and efficient data exchange.
- Collaborated on web-based dashboard integration for remote monitoring and control of embedded systems.

DUAL LIFE SCIENCE PVT. LTD.

06/2026 – Present

Electronics Engineer (Research & Development Engineer)

- Design, develop, test, and troubleshoot electronic circuits and embedded systems for R&D applications.
- Develop and integrate microcontroller-based systems with sensors, actuators, and communication interfaces.
- Work with ESP8266, ESP32, STM32, Arduino UNO, Raspberry Pi, and embedded platforms for IoT and real-time monitoring applications.
- Implement and troubleshoot communication protocols including I2C, SPI, UART, and CAN.
- Perform hardware testing, debugging, circuit validation, and component-level troubleshooting.
- Support prototype development, system integration, and performance optimization of electronic products.
- Collaborate with the R&D team to develop reliable and innovative hardware-software solutions.

PROJECTS

Smart EV Charging Controller using ESP8266 (IoT-Based System)

Tech Stack: ESP8266 NodeMCU, Embedded C, SPI, I2C, MFRC522 RFID, DS3231 RTC, ACS712 Current Sensor, EEPROM, HTTP Dashboard

- Developed an EV charging controller managing authentication via RFID and session tracking via RTC.
- Integrated local EEPROM data storage for user logging alongside an HTTP-based Admin Dashboard & User Portal.

Mecanum Wheel RC Car

ESP32, Motor Drivers, Arduino IDE, Robotics

- Designed an omni-directional RC car controlled wirelessly via smartphone over an ESP32 microcontroller with precise motion algorithms.

Radar System

Ultrasonic Sensor, Servo Motor, Arduino, Processing

- Created an object detection radar mapping a 180-degree field and visualized proximity data via a real-time graphical interface in Processing.

Alcohol Detection System & Home Automation Smart Gadget

Arduino, MQ-3, Blynk App, NodeMCU

- Built vehicle safety thresholds triggers with MQ-3 sensors and designed Wi-Fi appliance control interfaces via the Blynk IoT platform.

TECHNICAL SKILLS

Languages:	C/C++, Python, HTML, Assembly
Protocols / HW:	I2C, SPI, UART, RS-485, USART, Wi-Fi/Bluetooth Hardware Integration
Frameworks:	Embedded Systems, PCB Design, Arduino IDE, MATLAB, PLC, VLSI, Satellite Communication
Tools & Soft:	Git, VS Code, PyCharm, Microchip Studio Teamwork, Problem-Solving, Adaptability, Time Management

EDUCATION

Government Engineering College, Modasa

08/2022 – 06/2026

Bachelor of Engineering (B.E.), Electronics and Communication Engineering • Modasa, Gujarat

SGVS High School

05/2019 – 04/2021

Higher Secondary Certificate (HSC) • Surat, Gujarat

CERTIFICATIONS (LERNX – APRIL 2023)

Project Management (ID: ab8lmapldri)

- Covered agile/scrum frameworks, resource planning, budgeting, and project risk mitigation.

Python Programming (ID: python-gaof80f89w8)

- Mastered OOP, fundamental data structures, data scrapers, and analysis syntax.

MS Excel Certification (ID: iuoftahh28)

- Advanced formulas, pivot tables, data visualization, and business dashboards.